AI Programming Foundations Project

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Capstone: Advanced Applied Computer Science

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Overview

This project presents a complete, reproducible data workflow using Python in the form of Jupyter Notebook. The notebook contains all the necessary logic for cleaning and exploring data in a form of easy to use functions. The dataset used for this project - *Titanic - Machine Learning from Disaster* - was collected from Kaggle from the following URL: https://www.kaggle.com/c/titanic/data?select=train.csv

Dataset Description

Titanic - Machine Learning from Disaster dataset contains 891 rows and 12 columns. Each row represents one passenger of the Titanic final voyage. Data contains information about passengers age, sex, ticket class and number, fare, cabin, port of embarkation and number of family members on board. The column of most interest is called Survived and indicates if a given passenger survived the disaster.

Workflow Description

Project workflow consists of several steps. First data is loaded into DataFrame, then some initial inspections are performed to discover data quality issues. Next, using insights from previous step cleaning functions are defined and then applied on data to handle missing values, transform columns and remove redundant variables. After that exploratory data analysis is performed exploring particular features as well as relations between them. Exploration ends with

visualizations presenting interesting patterns in data. The whole project culminates with a final summary and findings.

Key Decisions and Assumptions

Data preprocessing is a crucial part of any data related project (Danchev, 2022). Decisions on how to clean the dataset impact any further steps in the data pipeline. In this project columns that required most attention were ones containing name, age and cabin number. Column with cabin numbers had high cardinality with the highest number of missing values. It was impossible to fill missing values with any reasonable defaults, because of that only deck identifiers were extracted from cabin numbers, limiting cardinality, and for missing deck info value X was used. Names are natural personal identifiers and because of that they have very high cardinality and couldn't be used directly in analysis in a way other than as labels. To get the most of the limited data, titles included with names were extracted and rare cases grouped. This approach enabled the use of part of the information contained in the Name column in a meaningful way in further analysis. When it comes to age, missing values were filled with the median of known age for groups defined by ticket class, sex, deck and title - assumption here being that people with most similar other features would most likely be also similar in respect to age. For example on average one would expect women titled mrs to be older than ones titled miss.

Exploratory data analysis which followed data cleaning focused on understanding variables present in the dataset, as well as relations between them. That was also the goal of produced plots - visualizations displayed how categorical variables related to survival, presented

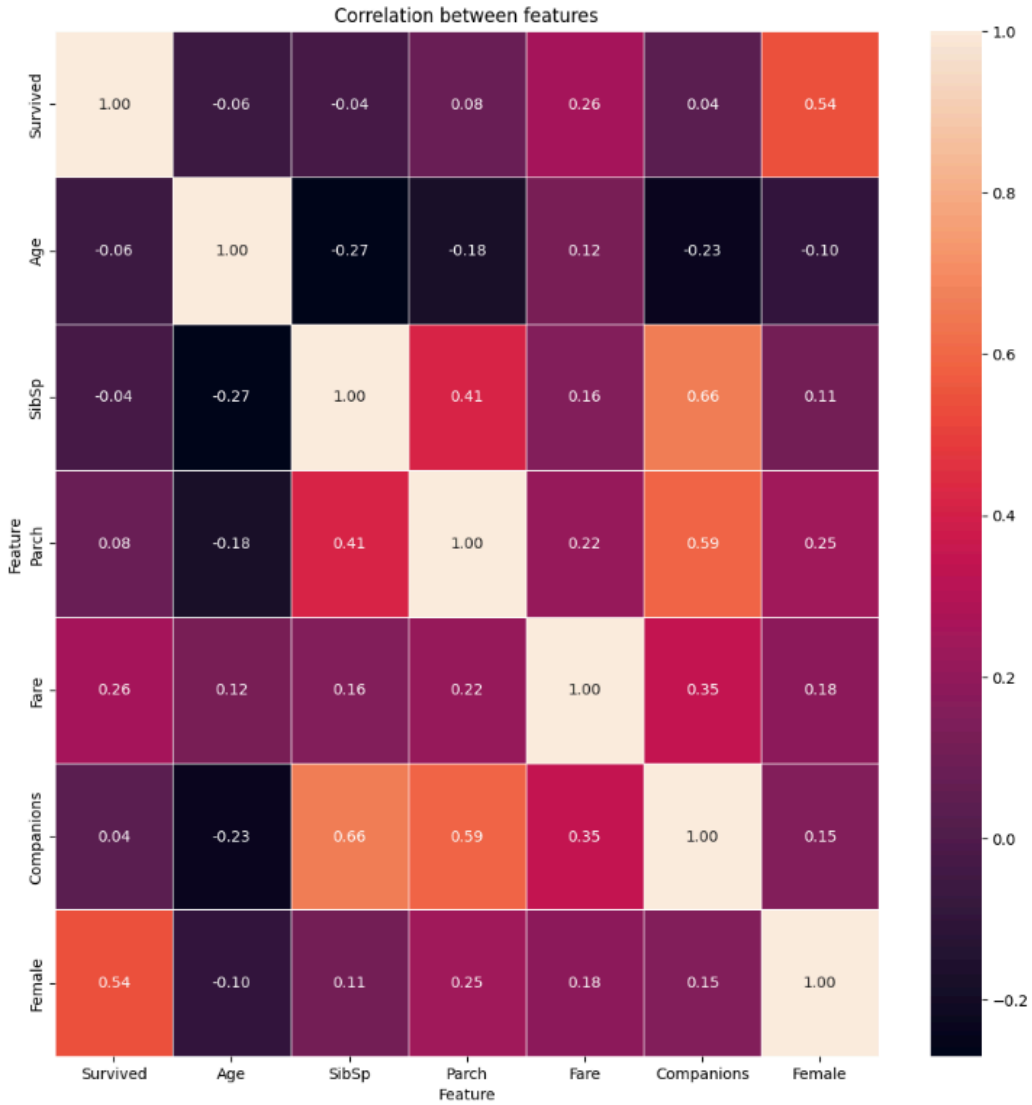
correlation in a graphical way and showed relations between selected features and survival. Assumption here being that survival was the most interesting way to explore the dataset.

Results and Interpretation

Investigation of relations between columns, in particular Pearson correlation, showed that there were no features strongly correlated with survival. Sex (being female) had the strongest correlation with survival among all columns, but the correlation was only moderately strong - with a value of 0.54 - as can be seen in Figure 1. This shows that even though most of the survivors were female (as visible in Figure 2.), there were other factors that influenced the outcome, and that influence wasn't necessarily linear.

Figure 1

Correlation between numeric features

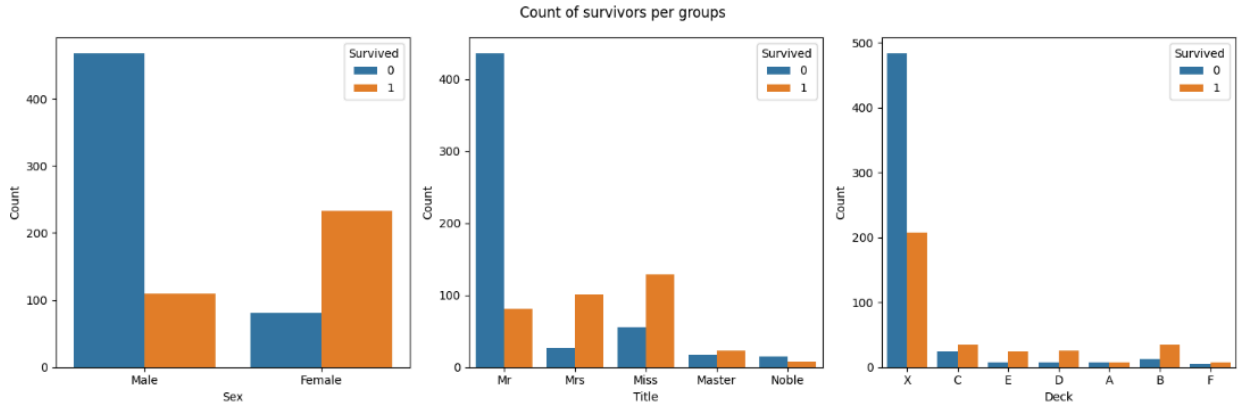


When it comes to categorical variables, their relation to survival was investigated by calculating and visualizing how many passengers from each group survived the tragedy. As can be seen in Figure 2. deck is an interesting variable - in groups when deck was known (cabin number was present) more people seem to have survived. If rows with missing deck had been dropped, that may have introduced a form of survivorship bias (Elston, 2021) - as it seems that

deck is known mainly for passengers who have survived and who thus were able to later report on their lodgings on the ship. When it comes to the titles, they seem to just strengthen the message already conveyed by plot for sex - that passengers prioritized saving females.

Figure 2

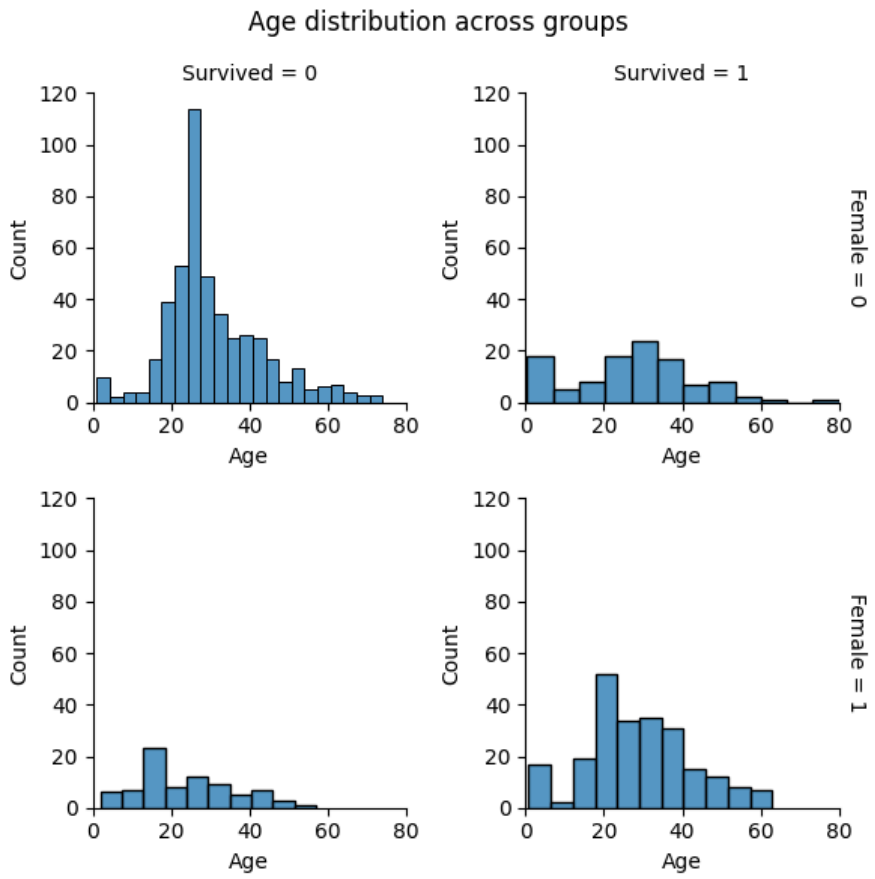
Count of survivors per different groups



Once sex is crossed with age - as can be seen in Figure 3. - it shows that the highest number of casualties were males of around 20 years of age. Distributions may not be exact due to missing values present in the original dataset, but the general outline shows that passengers tried to prioritize saving women, children and older people.

Figure 3

Age distribution across groups defined by survival and sex



Responsible Practice (Bias and Data Quality)

As mentioned in the previous section, it seems that if all rows with missing values were simply dropped, that may have introduced a form of survivorship bias, which can be seen for example when looking at the deck variable (present in Figure 2.). It appears that missing values were mainly present in rows representing passengers who did not survive, if such rows were to be dropped influence of other features may have been exaggerated and the general insights skewed.

To improve quality of the dataset in this case, the best approach seem to be consulting external sources such as Encyclopedia Titanica, as topic of Titanic disaster had been studied for a long time, and some missing information could be possibly accurately filled instead of inputting them with simple heuristic method.

Reproducibility

The whole project can be reproduced by following instructions and using contents present in the GitHub repository (https://github.com/patryk126p/ai-programming-foundations-project). Every change in the repository contents can be inspected allowing for reproducing the whole project progress. No manual data manipulation was performed which is one of the pillars of reproducible projects (Sandve, 2013). What's more, file *requirements.txt* allows to recreate the exact Python environment used for running the project. The final piece of reproducibility is the dataset used in this project included in the GitHub repository as well. At no point were there used any random or pseudo-random variables, which additionally ensures that project code runs exactly the same each time.

References

Danchev, V. (2022). Reproducible Data Science with Python: An Open Learning Resource. Journal of Open Source Education. 5. 156. 10.21105/jose.00156.

Sandve, G. K., Nekrutenko, A., Taylor, J., & Hovig, E. (2013). Ten simple rules for reproducible computational research. PLOS Computational Biology, 9(10), e1003285. https://doi.org/10.1371/journal.pcbi.1003285

Elston, D. M. (2021). Survivorship bias. Journal of the American Academy of Dermatology.